

Framing Effect, Risk Taking, and Ambiguity Aversion: A Coin Flipping Task

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Abstract (Written and edited by: Lina Robayo)

This study investigates how framing effects influence risk-taking and ambiguity aversion in decision-making. Grounded in Prospect Theory, we hypothesize that individuals exposed to positive framing (emphasizing gains) will exhibit greater risk aversion and prefer options with known probabilities, whereas those exposed to negative framing (emphasizing losses) will display increased risk-seeking behavior and a higher preference for ambiguous options. Participants (N = 60) will complete a 20-trial web-based coin-flipping task in which they repeatedly choose between a coin with a known probability and one with an unknown probability, both yielding equal expected value. Following the task, participants will report their satisfaction using a Likert-scale survey. We expect that individuals in the negative-framing condition will select the ambiguous coin more frequently and report lower satisfaction, despite similar monetary outcomes across conditions. These findings would suggest that framing significantly shapes decision preferences independently of objective value, highlighting the psychological impact of how information is presented. Understanding these effects has important implications for real-world decisions in domains such as healthcare, finance, and public policy, where outcomes are often uncertain and subject to interpretation.

Introduction (Written and edited by: Brielle Lauber)

The decisions we make every day are influenced by risk and uncertainty. They require us to evaluate the value of potential outcomes to our choices. To understand why people make the decisions they do, it is important to consider expected utility and the Prospect Theory, which work to explain why certain risks are taken while others are avoided. The Expected Utility Theory is a normative model typically used to explain rational choice. It describes how a reasonable person should choose between risky choices by using the expected value of the outcome to maximize gain (Kahneman & Tversky, 1979). Although this theory makes logical sense, it is often not the initial strategy used in decision-making. Furthermore, the Expected Utility Theory has been critiqued by the Prospect Theory, which is a descriptive model used to explain how people tend to make decisions in the real world. It suggests that individuals tend to evaluate outcomes in terms of changes relative to a reference point (Levy, 1992).

People tend to weigh losses more heavily than equivalent gains, and do not think in terms of net assets. These patterns lead to loss aversion (Levy, 1992).

The way people evaluate their choices and the probability of loss can be influenced by the way information is presented. This is a pattern derived from the Prospect Theory called the Framing Effect, in which choices with the same expected value, when framed in different ways, can change an individual's preference (Tversky & Kahneman, 1981). This tendency can have implications when individuals are making decisions involving risk. One example of a context in which the Framing Effect has been seen to affect responses is when rating medical treatments. For example, it has been shown that framing medical procedures as having a "50% success rate" rather than a "50% failure rate" can lead individuals to evaluate the treatment as more effective. It also increases the likelihood that they recommend it to others despite the same value being presented (Levin et al., 1987). The Framing Effect explains that people tend to avoid risk when a decision is framed as a "gain" and are more inclined to take risk when the task is framed as a "loss" (Seo et al., 2010).

If uncertainty cannot be reduced even when new information is gathered, it is considered aleatory and falls under the category of risk. On the other hand, epistemic uncertainty can be reduced by seeking out new knowledge and falls under the category of ambiguity (Fox et al., 2021). Although many studies have examined the framing effect on known probabilities, less work has examined how the framing effect and ambiguity influence choices in a repeated decision-making paradigm. In this study, we use a skewed coin-flipping task under positive and negative framing conditions to test if variations in the context in which information is presented influence participants' aversion to a coin with an unknown probability of winning. We aim to answer whether framing the probability of earning money positively or negatively influences participants' risk preferences. We hypothesize that those in the positive-framing condition will be more likely to choose the coin with a known probability, showing greater risk aversion. We also hypothesize that those in the negative-framing condition will be less likely to choose coins with a known probability, showing less risk aversion. Finally, we hypothesize that those in the negative-framing task will report lower satisfaction with their performance. This work is important because understanding how framing

influences decisions can provide insight into why people make the decisions they do about healthcare, politics, finances, careers, housing, and any other daily considerations, where outcomes are uncertain.

Methods (Written and edited by: Praneeth Puli)

The experimental task consists of a web-based coin-flipping simulation where participants complete 20 discrete trials. In each trial, participants are presented with two virtual coins and must choose one to flip, with the primary objective of accumulating as much money as possible in a "bank" that starts at \$0. The first option, Coin A, presents a known probability with an explicitly skewed outcome with either a "40% chance to win \$15" or a "60% chance to lose \$7." On the other hand, Coin B represents the ambiguous choice, meaning the specific percentages are hidden from the participant to create epistemic uncertainty, but it still has the same fixed probability. To ensure that choices are driven by psychological framing rather than objective logic, both coins are programmed with the same expected value (EV). When completing the 20 trials, participants are shown their total earnings and asked to complete a 5-point Likert-scale survey to assess their performance satisfaction relative to their initial expectations.

The experimental software was custom-built using a combination of web technologies and cloud-based automation to ensure seamless data collection and participant tracking. The coin-flipping simulation was developed as a web-based application using HTML, CSS, and JavaScript, with the core logic and scripts generated by Gemini. GitHub was utilized as the primary repository for version control and collaborative development, allowing for testing of the randomizing algorithms and the user interface. To automate data logging, the application integrated a Google Apps Script backend that transmitted participant responses including coin choices, trial outcomes, and total earnings in real-time via HTTPS requests directly into a centralized Google Sheets database.

This study consists of two study group designs to isolate the specific effects of framing on decision-making. The independent variable is the Framing Condition, which is manipulated at two levels which are positive framing, where outcomes are emphasized as gains, and negative framing, where the same outcomes are presented

as potential losses. The study measures three dependent variables. These are ambiguity preference (the percentage of trials in which a participant selects the ambiguous coin), performance satisfaction (derived from the post-task Likert-scale score), and total earnings (recorded as a control variable).

We plan to recruit a total of 60 participants ($N = 60$), with 30 individuals randomly assigned to each condition. This sample size satisfies the central limit theorem and is justified by a power analysis to conduct an independent-samples t-test. With an alpha level of 0.05 and a desired power of 0.80, a sample of 60 is sufficient to detect a medium-to-large effect size, consistent with the shifts in risk preference historically observed in framing research (Levin et al., 1988).

Expected Results (Written and edited by: Praneeth Puli)

We predict that framing will significantly influence participants' risk preferences, even when the expected value of the coins is identical across all trials. Based on the Prospect Theory, individuals do not process gains and losses equally. Specifically, we expect participants in the negative framing condition to make more choices involving the ambiguous coin compared to those in the positive framing condition. This would show a clear shift toward risk-seeking behavior when a choice is presented as a potential loss, as participants are more likely to "gamble" on an unknown probability to avoid a certain negative outcome. Across the 20 trials, we expect the mean number of ambiguous coin selections to be significantly higher in the negative-framing group than in the positive-framing group. In the positive frame, we anticipate a strong preference for the coin with the known probability reflecting risk aversion since participants seek to secure a "guaranteed" win. On the other hand, the negative frame is expected to cause ambiguity seeking, where the unknown probability of the second coin is perceived as the better choice than a clearly defined loss.

To test these different predictions, we will perform an independent sample t-test to compare the number of ambiguous coin decisions made under the two conditions. Assuming that our hypotheses are correct, the group that was given a negative frame will yield a significantly greater mean value of the decisions, giving a p-value less than 0.05. We will also analyze the total monetary earnings between the two groups. Given

that the expected values are equal, there will be no significant differences in the money earned. This control procedure will help us conclude that the effects of framing manipulation resulted purely from the framing manipulation process. Finally, we believe that the effects of framing will influence the subjects' satisfaction with their outcomes. In other words, we predict that despite the similarities in terms of the money earned, the subjects in the negative framing condition will have a satisfaction at a significantly lower level of Likert scale questions and vice versa for the positively framed subjects.

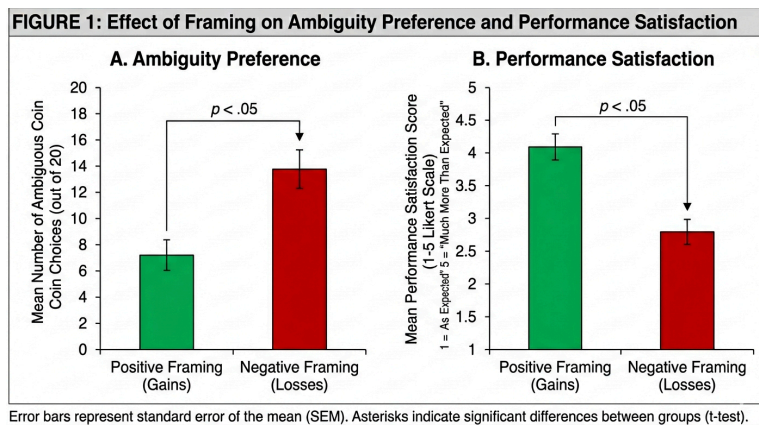


Figure 1. Hypothesis-Driven Visualization of Framing Effects. Panel A: Compares mean ambiguous choices; illustrates the predicted shift toward risk-seeking in loss frames with a hypothetical $p < .05$ significance marker. Panel B: Compares satisfaction scores; depicts the initial hypothesis that negative framing would significantly lower performance satisfaction. Error Bars: Represent the Standard Error of the Mean (SEM).

Results & Discussion (Written and edited by: Praneeth Puli)

From the data reported it reveals that framing conditions influenced behavioral preferences toward ambiguity, but individual variability was high. Participants in the negative framing condition ($n=10$) selected the ambiguous coin an average of 14.20 ± 3.61 times, while those in the positive framing group ($n=17$) on average chose it 11.65 ± 6.17 times. Although an independent samples t-test showed that this difference did not reach traditional statistical significance ($t(25) = 1.19$, $p = .245$), the moderate-to-large effect size (Cohen's $d = 0.50$) suggests a meaningful trend toward ambiguity seeking when outcomes are presented as losses. This behavior aligns with the Prospect Theory, where a negative frame likely triggers amygdala activation, biasing the ventromedial

prefrontal cortex (vmPFC) to favor uncertain options in an attempt to "gamble" away from a certain negative reference point(De Martino et al., 2006).

Regarding subjective outcomes, the negative framing group reported a higher mean satisfaction rating (2.90 ± 1.10) than the positive framing group (2.18 ± 1.13). This unexpected result yielded a large effect size (Cohen's $d = 0.65$) and a significance of ($t(25) = 1.63$, $p = .115$), contradicting the hypothesis that negative framing would lower satisfaction. This trend may be explained by the resolution of epistemic uncertainty. The anterior insula typically tracks the discomfort associated with ambiguity, and navigating this state to achieve a positive result can trigger dopamine release in the ventral striatum(Huettel et al., 2006). Participants who "gambled" on the ambiguous coin and avoided the "sure loss" may have experienced a higher subjective sense of achievement, despite the identical expected values across both conditions(Fiorillo et al., 2003).

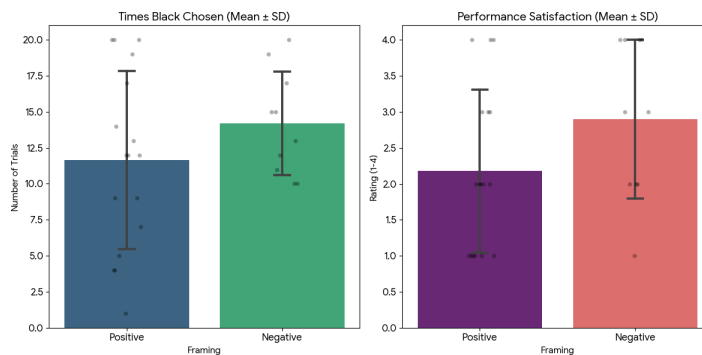


Figure 2. Behavioral Results of the Coin Flipping Task (Mean $\pm$ SD). Left Panel (Ambiguity Preference): Shows average trials selecting the ambiguous coin. Right Panel (Performance Satisfaction): Displays 1–4 Likert satisfaction.

Neuroscience Extension (Written and edited by: Praneeth Puli)

To investigate the neural mechanisms underlying the framing effect and ambiguity aversion, we would most likely use functional magnetic resonance imaging (fMRI). Trial timing would be adjusted to include a "jittered" inter-trial interval (ITI) of 2 to 6 seconds. This temporal spacing is necessary to allow the blood-oxygen-level-dependent (BOLD) signal to return to baseline between choices, allowing to isolate the neural activity associated with the decision-making phase from

the outcome phase. We would target multiple regions identified in previous neuroeconomic research. This includes the amygdala, known for its role in emotional processing, particularly in response to potential losses. Research by De Martino et al. (2006) demonstrated that amygdala activity is specifically associated with the framing effect, suggesting an emotional system modulates these decisions. Secondly, the ventromedial prefrontal cortex (vmPFC) is involved in calculating and comparing the subjective value (utility) of different options. Tom et al. (2007) identified the vmPFC as a key site for representing "neural utility," where activity levels shift based on the magnitude of potential gains versus losses. Lastly, the anterior insula is associated with disgust or unfairness signals and the processing of uncertainty. Huettel et al. (2006) found that decision-making under ambiguity (epistemic uncertainty) selectively activates the anterior insula, distinguishing it from the neural processing of known risks.

With our hypotheses, we predict that participants who exhibit a high susceptibility to the framing effect will show significantly increased amygdala activation during negative framing compared to positive framing. This would support the idea that the negative framing effect is driven by an emotional "affective" response to the prospect of loss, supported by De Martino et al. (2006). We also predict that choosing the ambiguous coin will correlate with heightened activity in the anterior insula, particularly in the positive framing condition where the known coin is perceived as a safe gain. This aligns with findings that the insula tracks the discomfort or "disgust" associated with missing information in ambiguous contexts (Huettel et al., 2006). Lastly, we predict that participants who chose the "known" coin even in a negative frame because the EV is equal will show increased activity in the anterior cingulate cortex (ACC) and dorsolateral prefrontal cortex (dlPFC). According to the findings of De Martino et al. (2006), this would suggest that higher-order cognitive control is required to override the initial emotional bias triggered by the negative frame. We expect the vmPFC to act as a common scale for representing the subjective value of both the risky and ambiguous options (Levy et al., 2010). By correlating BOLD signal intensity with the behavioral choice to pick the risky/ambiguous coin, we can determine whether the "Loss" frame truly shifts the neural representation of value or simply increases the emotional cost of selecting a "certain" outcome.

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